The WATERS Network: *An Integrated* Environmental Observatory Network *for* Water Research

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An integrated real-time distributed observing system could transform the understanding of the earth's water and related biogeochemical cycles across multiple spatial and temporal scales to enable improved forecasting and management of critical water processes affected by human activities.

Water is vital to our well-being, but this critical resource is stressed by multiple and sometimes competing demands—human consumption, agriculture, industry, recreation, electric power, and ecosystem requirements. For example, the Chesapeake Bay, despite years of federal, state, and local restoration efforts, has shown only modest improvements in water quality since 1983 (*1*). Litigation, under way or threatened, has erupted in the eastern U.S. over water allocation (e.g., over the Apalachicola system among Georgia, Florida, and Alabama and over the Potomac River between Virginia and Maryland). In recent years, when faced with rewriting the master manual for the Missouri River, the U.S. Army Corps of Engineers had to consider not only navigation but also hydrologic regimes for maintaining habitats for fish and migratory birds (*2*). Western states experiencing increasing population densities coupled with diminishing seasonal snowpacks are concerned that the current water distribution infrastructure will soon not meet their growing demand. Communities across the country are considering reusing treated wastewater to augment existing freshwater supplies.

To address this growing crisis, the existing engineering and scientific knowledge base must be enhanced to provide improved input to decision makers. Currently, water is treated as a fragmented resource. For example, water quality is separated from water quantity and groundwater from surface water. Scientists, engineers, and policy makers are hampered in their ability to analyze and forecast environmental change, because they lack sufficient knowledge of the dynamics and spatial variability of relevant environmental processes and of how long-term environmental phenomena and human activities interact to influence these processes. A new joint initiative of the environmental engineering and hydrologic sciences research communities, with the support of the National Science Foundation (NSF), is proposing to build a national platform for real-time environmental observations with distributed networked sensing, high-performance computing, and data mining techniques. Once in place, the Water and Environmental Research Systems (WATERS) Network will enable multiscale research and experimentation on U.S. water resources.

The fundamental premise of the WATERS Network is that knowledge of the physical, chemical, and biological mechanisms controlling water quantity and quality is limited by lack of observations at the necessary spatial density and temporal frequency needed to infer the controlling processes. Furthermore, these measurements must be made at the right time and place; that is, hot spots and hot moments occur that control observed patterns disproportionately to their extent or duration. Traditional environmental sampling is too sparse and too infrequent to capture these processes.

History of the WATERS Network

During the past decade, the environmental science research community, with the support of NSF, has been developing the vision for numerous environmental observing systems, including EarthScope, the National Ecological Observatory Network, and the Oceans Observatory Initiative. In its 2001 report, *Grand Challenges in Environmental Sciences*, the National Research Council identified biogeochemical cycles, biological diversity and ecosystem functioning, climate variability, hydrologic forecasting, infectious diseases and the environment, institutions and resource use, land-use dynamics, and reinvention of the use of materials as the eight grand environmental challenges facing the U.S. The environmental engineering research community responded to this call by beginning discussions on the need for a nationwide environmental observatory and research initiative focused on human-induced impacts on the environment. Similar discussions were also taking place in the hydrologic science research community about the need for an observatory initiative focused on natural processes. The WATERS Network was developed in 2005 as a combination of these two national planning initiatives—CLEANER (Collaborative Large-Scale Engineering Analysis Network for Environmental Research) and the CUAHSI (Consortium of Universities for the Advancement of Hydrologic Science) initiative for hydrologic observatories.

Vision for the WATERS Network

The U.S. has a long history of collecting data on traditional water-quality and water-quantity parameters (e.g., the U.S. Geological Survey [USGS] National Water Information System [NWIS], the U.S. EPA National Assessments Database, and the National Pollutant Discharge Elimination System permitting requirements). Unfortunately, much of this information is fragmented, site-specific, and not easily accessible to researchers interested in investigating multiscale environmental phenomena. To address this deficiency, the WATERS Network plans to de-

ploy an integrated and distributed system of environmental observatories at sites across the country. The network will cover a range of spatial scales and climate and land use/land cover conditions and focus primarily on human-dominated landscapes. Individual sites may be compared either hierarchically as nested watersheds (from plot-scale to basin-scale) or across a single scale as "problemsheds". Research at these sites would be aided by tools for collection, storage, and dissemination of environmental data; interactive models that could be tested in real or near-real time; and an integrative cyberenvironment that would help multidisciplinary, geographically dispersed teams of researchers work together effectively. The network will complement and leverage the existing investment in water monitoring operated by various federal and state agencies, river basin commissions, and other entities.

The WATERS Network will address the following key scientific questions. How are water quantity, quality, and related earth system processes coupled to natural and human-induced changes in climate and the environment? How do engineered systems process and modulate water and water-borne constituent fluxes across various scales and in different settings and climate regimes? How can engineered and socioeconomic systems and institutions adapt to changes in water quantity and quality? How can new science and technology be used to design water-resource systems and policy instruments and promote sustainable solutions to problems of water quantity (scarcity or excess) and quality?

The network will enable researchers to conduct multiscale, dynamic, predictive modeling for water, sediment, and water quality, including point- to national-scale prediction capability; real- or near-real-time assimilation of data, where appropriate, for transient phenomena or process feedback; and feedback for observatory operation in response to events. A fractional factorial design will be used to yield a workable number of sites that will allow testing of theories and technologies across the range of conditions nationwide. The master design variables include climate, geographical location, land use/land cover, population density, land form, hydrologic setting, and scale.

At this time, we are proposing that the common data to be measured across the network include water balances, flow paths and rates, sediment transports, and basic water-quality parameters (e.g., temperature, suspended solids, salinity, pH, and dissolved oxygen). Additional parameters (depending on specific observatory location and science needs) include nitrogen, phosphorus, carbon, *E. coli*, and enterobacter concentrations. New sensors will be incorporated into the network as they develop. We anticipate that these data will permit transformative research on basin-scale transport of water, sediments, nutrients, and pathogens and will provide a strong platform for research studies that require more manually intensive data collection. For most of these systems, deploying a tiered or multiscale sensing network is integral to achieving the observational resolution required to respond to the above goals. Such an approach would rely on the use of existing satellite-based sensors and strategic deployment of new ground-based sensor networks coupled with data from relevant existing networks (e.g., USGS NWIS). Ideally, the ground-based sensors would be capable of multivariable measurements and suitable for extended and unattended deployments in the field. Atmospheric, terrestrial, and subsurface sensor technologies that have been used successfully in other fields (e.g., agronomy, plant and soil ecophysiology, hydrogeology, meteorology, and geophysics) also will be integrated into the network.

A multiscale observational network with a combination of stationary and mobile sensors is being tested on the San Joaquin River in central California to identify key salinity-impacted flow conditions, which will enable scientists to better manage salt loads (box on p 6645). This project focuses on quantifying spatially distributed hydraulic and water-quality parameters. An alternate sensing approach with human-controlled and autonomous robotic devices for sensor deployment in rivers (e.g., autonomous underwater vehicles [AUVs] or vertical profilers) would enable spatially continuous sampling in multiple dimensions. The box on p 6646 describes how researchers at Rensselaer Polytechnic Institute (RPI) are using AUVs equipped with sensors for high-resolution 3D mapping in water bodies. The lessons learned from these and other, related field projects will be used in the design of the WATERS Network.

Cyberinfrastructure

The ability of the network to address grand challenges depends on the implementation of critical cyberinfrastructure. This platform is expected to include high-performance computing tools and intensive database management for the collection, storage, and dissemination of environmental data; advanced visualization tools; community-vetted models for system and process synthesis that can be used in near-real time; and collaboration and knowledge networking tools that will help multidisciplinary and geographically dispersed teams of researchers to work together effectively. Research is already under way on the feasibility and application of several of these technologies. For example, the CUAHSI Hydrologic Information System (HIS; www.cuahsi.org/his.html) is developing a geographically distributed set of tools and data sets that allows user access to hydrologic information from federal and other public sources. The HIS web portal is available for querying and downloading surface-water, groundwater, and water-quality data from the USGS NWIS, water-quality data from EPA, climate data from the National Climatic Data Center, remote-sensing information from NASA, and weather data from the National Center for Atmospheric Research and from Unidata.

Other research on workflow integration technology and tools for collaboration and knowledge networking is being conducted by the National Center for Supercomputer Applications' Environmental CyberInfrastructure Demonstration (ECID) team at the University of Illinois Urbana–Champaign (http://

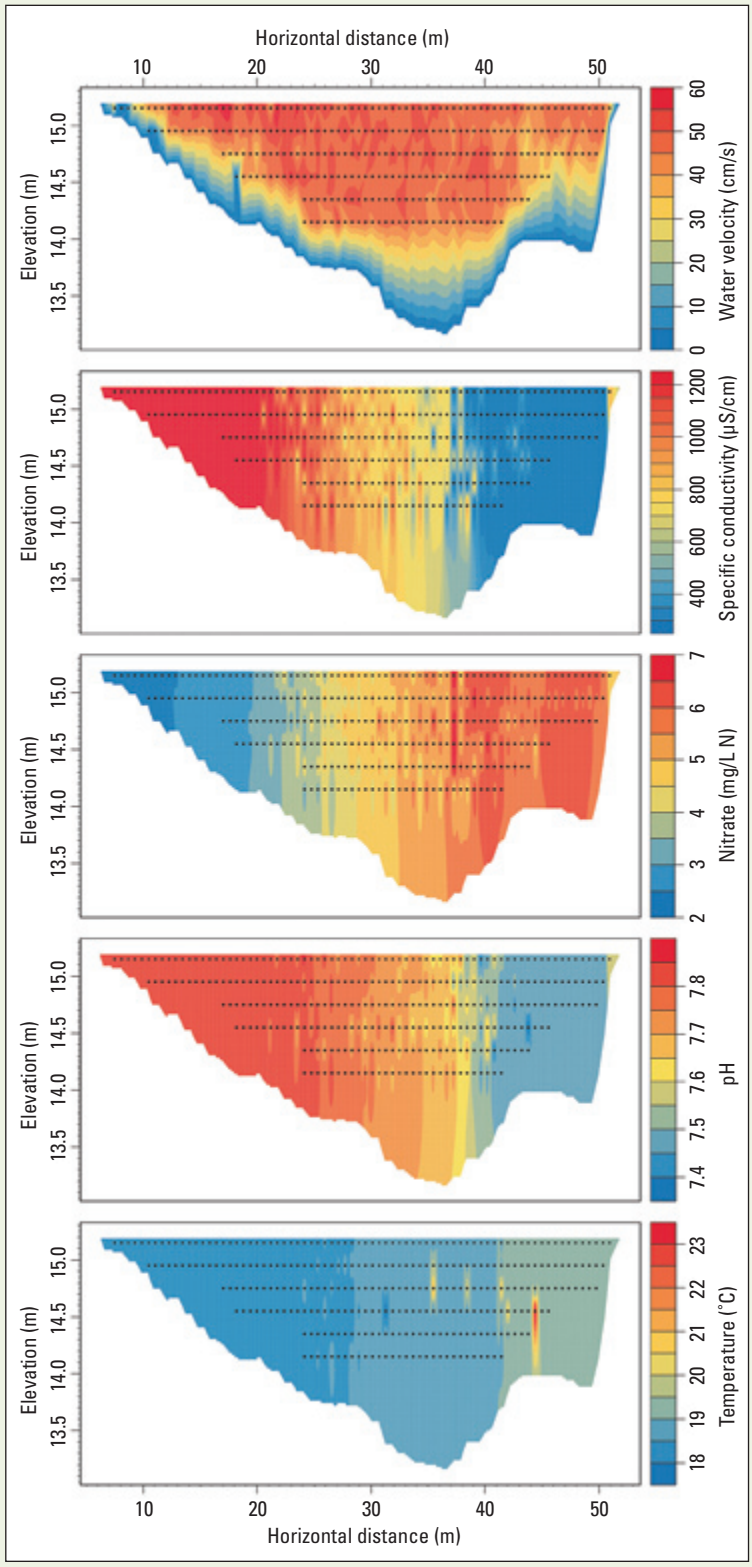
High-resolution river observations in central California

Researchers from the University of California Merced and University of California Los Angeles's NSF Center for Embedded Networked Sensing are testing a multiscale observational network on the San Joaquin River in central California. They ask the question: how can we better understand and manage salt loads on the San Joaquin River? This project, originally developed with funding from the NSF Information Technology Research Program and the CLEANER initiative, is now part of a larger scale WATERS Network virtual hydrologic observatory development project funded by NSF. A novel robotically manipulated sensing technology enables automation of river observations at an unprecedented level of spatiotemporal resolution. The Rapidly Deployable Networked Infomechanical System (NIMS RD) provides reproducible delivery of hydrodynamic and water-quality sensors to preprogrammed or adaptively identified sampling locations within a river cross section (*7*).

This demonstration project focuses on quantifying spatially distributed hydraulic and water-quality parameters just downstream of the confluence of the agriculturally impacted Merced and San Joaquin rivers. The San Joaquin River basin is equipped with synoptically sparse, networked gauging stations (separated by tens of kilometers), which provide time-series data on river stage, flow, salinity, and temperature in real time via the Department of Water Resources' California Data Exchange Center (CDEC). CDEC data allow investigators to identify key salinity-impacted flow conditions upon which to focus human-actuated river surveys and stationary sensor placement to characterize the bathymetry and water quality at key river segments. This information facilitates optimal deployment of the NIMS RD sensor platform. Figure 1 shows the coupled cross-sectional velocity and water-quality parameters distributions for a 55 meter span of the San Joaquin River. The data reveal concentration gradients at previously unobservable resolution, which, when coupled with the observed velocity field, result in improved estimates for a distributed chemical flux. Reproducible, distributed observations such as these allow for the quantitative assessment of environmental quality as a function of human impacts (here, agricultural practices and reservoir operation). In the future, networked sensing data, such as those from CDEC, will be used to continuously calibrate computational river models to enable real-time forecasts for triggering and optimizing deployments such as those described here.

FIGURE 1

A sample of simultaneously collected NIMS RD data for a cross section located downstream of the San Joaquin–Merced confluence (top to bottom): velocity, specific conductance (salinity), nitrate, pH, and temperature.



isda.ncsa.uiuc.edu/ecid/intro.html). The ECID team is developing a prototype cyberenvironment that couples traditional desktop computing environments with the resources and capabilities of a national cyberinfrastructure to provide researchers with the ability to access, integrate, automate, and manage complex, collaborative projects across disciplinary as well as geographical boundaries. The prototype cyberenvironment includes a workflow integration technology called CyberIntegrator, a collaboration system called CyberCollaboratory, and a knowledge networking service called CI-KNOW. CyberIntegrator allows heterogeneous workflows (or tasks) and software tools—often created by different users with multiple software technologies—to be linked and executed within one user-friendly system. The CyberCollaboratory (http://cleaner.ncsa.uiuc.edu/cybercollab) is a collaborative environment in which communities of researchers and other stakeholders can share knowledge and information, analyze data, and solve problems collaboratively. CI-KNOW provides referrals to users about resources related to their current activities, such as publications created from data that they are accessing.

Distributed sensor networks and environmental robotics in the Hudson River

Researchers at RPI, in conjunction with collaborators at Columbia University's Lamont-Doherty Earth Observatory, the New York State Department of Environmental Conservation, USGS, and the Beacon Institute for Rivers and Estuaries, are exploring the development of sensor networks for monitoring biological, chemical, and physical properties of environmental systems in the Hudson River and the estuary in New York. This study is part of an NSF-funded CLEANER planning grant (RiverScope: Large-Scale Engineering Analysis Network for Environmental Research on the Hudson River) and is exploring the organizations, infrastructures, and technologies needed to support new paradigms of scientific inquiry in a large-scale environmental system. Initial implementations of sensor systems have included deployment of multiple sensor nodes in the river and estuary observing salinity distribution, dissolved oxygen, suspended sediment, and chlorophyll. Related studies have focused on contaminant transport and migration of biological species. Initial results from these activities demonstrate the need for the development of more integrated networking capability that links sensors to modeling, simulation, and visualization capabilities over extended geographic areas.

A special focus of this program is the use of autonomous underwater vehicles (AUVs) equipped with sensors for high-resolution 3D mapping of a water column. A solar-powered AUV (SAUV) has been developed to support mobile deployment of sensors (see photograph on p 6647). The SAUV integrates control and communication with efficient energy use and solar recharging for long-term deployment. The SAUV recently demonstrated 90 hours (h) of deployed time in a 5-day period, recovering data on physical and chemical parameters of a freshwater lake. It operates submerged for up to 8 h when fully charged and is rated to dive to 100 meters. It returns to the surface to recharge (6–8 h of daylight), and during recent endurance tests, the SAUV remained operational in the water for more than 20 days. It is fully autonomous and supported by onboard computing, navigation, and communications systems. Interactive mission planning software provides a user-friendly environment for vehicle programming.

Other recent SAUV experiments have included monitoring of dissolved oxygen in a tidal estuary and communications and control among three AUV systems. Research in environmental robotics poses fundamental algorithmic issues of mission planning and adaptive sampling to guide the deployment of these mobile-sensing nodes and the optimal estimation of parameters of distributed variable field models. In this program, these topics are approached by using information measures of prediction error to guide autonomous deployment of the SAUVs.

Integrating social science research and transforming environmental education

A design that fails to include key regulatory constraints and economic and social forces may result in the collection of data that are insufficient to address critical drivers (*3*). For example, to evaluate pollutant loadings in a watershed, engineers and natural scientists typically study the behavior of released contaminants. However, this approach often neglects to incorporate the effects of release behavior on pollutant loading (e.g., the use of garden fertilizers, herbicides, and pesticides by homeowners and manure and fertilizer application schedules adopted by farmers). The lack of such data can impact the accuracy of fate and transport predictions. The following key social science questions should be included in the network design (*4*). Which human actions influence the availability of resources and disturbance regimes across aquatic and associated ecosystems? How do human-induced alterations to the environment lead to changes in ecosystem services? How do those changes in ecosystem services then affect humans and human values?

Clearly, U.S. waters are affected by the daily actions of American citizens—but few know what to do to improve these water problems. Moreover, the gap between job openings and degrees granted in science, technology, engineering, and mathematics (STEM) fields is widening at a rapid pace (*5*, *6*). In 2005, the U.S. Government Accountability Office (*6*) reported to Congress that during the past decade the proportion of students obtaining degrees in the STEM fields has fallen even though overall college enrollment has increased steadily during the same time period. Reversing this trend is crucial for keeping the U.S. scientifically competitive. The WATERS Network will work toward this aim by transforming environmental education at all levels, engaging our citizenry in water science research and management, and addressing significant workforce issues in many fields of science and engineering. Once built, the network should improve the recruitment and retention of students in these fields by providing real-world research experience and access to socially relevant, state-of-the-art, interactive, Internet-based curricula at all levels of the educational system.

Future perspectives

With the construction of the WATERS Network, investigators will have an unprecedented opportunity to combine data from laboratory investigations and single field sites with data collected nationwide to collaborate with colleagues in real time on complex environmental research questions. We hope to begin

construction of the network in early 2012 with the expectation that it will become fully operational by 2016. During the past 3 years, NSF has funded several million dollars' worth of individual observatory test-bedding research and planning grants to evaluate and optimize field deployment of environmental sensors and supporting cyberinfrastructure. Further test-bedding opportunities for individual researchers are expected between now and 2012.

Once approved by Congress, funding for the actual construction of the network will be distributed over 4 years and will come from the NSF Major Research Equipment and Facility Construction (MREFC) Program (*8*), an agency-wide special account set up to pay for the acquisition, construction, and commissioning of major scientific infrastructure and equipment. The MREFC account is designed specifically to prevent these larger obligations from disproportionately impacting individual directorate research budgets; ~$150–250 million per year is allocated for this purpose. Although the MREFC program will not support any ongoing operation and maintenance once the network is built, the project office is actively exploring partnerships with outside organizations and agencies to minimize impacts on future NSF research funding and leverage existing investments in water monitoring.

ARTHUR SANDERSON, RPI

A solar-powered autonomous underwater vehicle is used for high-resolution 3D environmental observation. (Developed by Autonomous Undersea Systems Institute, RPI, and Falmouth Scientific, Inc.)

As the WATERS Network community develops its design, we will ask engineers and biophysical and social scientists from the broader research community to provide input on both the conceptual design and the integrated science and education plan (for more information, go to www.watersnet.org). We also anticipate a need for broader involvement by the community in the planning for the network in the coming years as the design evolves and matures (to keep apprised of developments related to the WATERS Network and to subscribe to the *WATERS Network News*, visit www.watersnet.org). Although 2016 might seem distant, a project of this magnitude requires careful planning to ensure both the best use of federal dollars and the effective integration of the anticipated network capabilities with existing data collection and monitoring efforts. Ultimately, the national imperative for better forecasting and management of U.S. water resources requires a bold departure from previous approaches to solve these complex societal problems. We believe that the WATERS Network mission embodies that bold departure.

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